5054 Homework2

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1 Basic Knowledge

1.1 Show that the log-likelihood function is concave in logistic regression

Consider a binary classification problem with n independent observations $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$, where $\mathbf{x}_i \in \mathbb{R}^d$ is the feature vector and $y_i \in \{0,1\}$ is the label. In logistic regression, the conditional probability of $y_i = 1$ given $\mathbf{x}_i$ and parameter vector $\boldsymbol{\theta} \in \mathbb{R}^d$ is modeled as

$$p(y_i = 1 \mid \mathbf{x}_i; \boldsymbol{\theta}) = \sigma(\mathbf{x}_i^\top \boldsymbol{\theta}) = \frac{1}{1 + e^{-\mathbf{x}_i^\top \boldsymbol{\theta}}}.$$

The log-likelihood function for the entire dataset is

$$\ell(\boldsymbol{\theta}) = \sum_{i=1}^n \left[y_i \log \sigma(z) + (1 - y_i) \log\left(1 - \sigma(z)\right) \right], \quad z = \mathbf{x}_i^\top \boldsymbol{\theta}$$

Using the identity $\log \sigma(z) = z - \log(1 + e^z)$ and $\log(1 - \sigma(z)) = -\log(1 + e^z)$, this simplifies to

$$\ell(\boldsymbol{\theta}) = \sum_{i=1}^n \left[y_i z - \log\left(1 + e^z\right) \right], \quad z = \mathbf{x}_i^\top \boldsymbol{\theta}$$

To establish concavity of $\ell(\boldsymbol{\theta})$, it suffices to show that each summand

$$\ell_i(\boldsymbol{\theta}) = y_i z - \log\left(1 + e^z\right), \quad z = \mathbf{x}_i^\top \boldsymbol{\theta}$$

is a concave function of $\boldsymbol{\theta}$, since the sum of concave functions is concave.

Define the scalar function $f_i(z) = y_i z - \log(1 + e^z)$ for $z \in \mathbb{R}$. Then $\ell_i(\boldsymbol{\theta}) = f_i(\mathbf{x}_i^\top \boldsymbol{\theta})$. Since $\mathbf{x}_i^\top \boldsymbol{\theta}$ is an affine (in fact, linear) function of $\boldsymbol{\theta}$, the composition $f_i(\mathbf{x}_i^\top \boldsymbol{\theta})$ is concave in $\boldsymbol{\theta}$ if $f_i(z)$ is concave in z.

Compute the first and second derivatives of $f_i(z)$:

$$f_i'(z) = y_i - \frac{e^z}{1 + e^z}, \quad f_i''(z) = -\frac{e^z}{(1 + e^z)^2}.$$

For all $z \in \mathbb{R}$, we have $e^z > 0$ and $(1 + e^z)^2 > 0$, so

$$f_i''(z) = -\frac{e^z}{(1 + e^z)^2} < 0.$$

Thus, $f_i(z)$ is strictly concave on $\mathbb{R}$. Consequently, $\ell_i(\boldsymbol{\theta}) = f_i(\mathbf{x}_i^\top \boldsymbol{\theta})$ is concave in $\boldsymbol{\theta}$ for each i.

Since $\ell(\boldsymbol{\theta}) = \sum_{i=1}^n \ell_i(\boldsymbol{\theta})$ is a sum of concave functions, it is concave. Therefore, the log-likelihood function of logistic regression is concave in $\boldsymbol{\theta}$.

1.2 Besides the gradient descent algorithm and Newton algorithm, state another algorithm which can be used to find MLE for logistic regression. Explain the details and its pros and cons compared to Newton algorithm.

BFGS (Broyden–Fletcher–Goldfarb–Shanno algorithm) is a quasi-Newton method that iteratively minimizes a smooth objective function without explicitly computing the Hessian matrix.The key idea of BFGS is to maintain an approximation $\mathbf{B}^{(k)}$ of the Hessian matrix $\nabla^2 f(\boldsymbol{\theta}^{(k)})$ using only gradient information from previous iterations. The algorithm proceeds as follows:

1. Initialize $\boldsymbol{\theta}^{(0)} \in \mathbb{R}^d$ and set $\mathbf{B}^{(0)} = \mathbf{I}$ (the identity matrix).
2. For iteration $k = 0, 1, 2, \ldots$ until convergence:
3. Compute the gradient: $\mathbf{g}^{(k)} = \nabla f(\boldsymbol{\theta}^{(k)}) = -\nabla \ell(\boldsymbol{\theta}^{(k)})$.
4. Solve for the search direction: $\mathbf{p}^{(k)} = -(\mathbf{B}^{(k)})^{-1}\mathbf{g}^{(k)}$.
5. Perform a line search to find a step size $\alpha_k > 0$ satisfying the Wolfe conditions.
6. Update the parameters: $\boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} + \alpha_k \mathbf{p}^{(k)}$.
7. Compute the displacement: $\mathbf{s}^{(k)} = \boldsymbol{\theta}^{(k+1)} - \boldsymbol{\theta}^{(k)}$.
8. Compute the gradient difference: $\mathbf{y}^{(k)} = \mathbf{g}^{(k+1)} - \mathbf{g}^{(k)}$.
9. Update the Hessian approximation using the BFGS formula:

$$\mathbf{B}^{(k+1)} = \mathbf{B}^{(k)} + \frac{\mathbf{y}^{(k)}(\mathbf{y}^{(k)})^\top}{(\mathbf{y}^{(k)})^\top \mathbf{s}^{(k)}} - \frac{\mathbf{B}^{(k)}\mathbf{s}^{(k)}(\mathbf{s}^{(k)})^\top \mathbf{B}^{(k)}}{(\mathbf{s}^{(k)})^\top \mathbf{B}^{(k)}\mathbf{s}^{(k)}}.$$

Compared to Newton's method:
PROS:

- It does not require computing the exact Hessian matrix, which for logistic regression is $\nabla^2 \ell(\boldsymbol{\theta}) = -\sum_{i=1}^n \sigma_i(1-\sigma_i)\mathbf{x}_i\mathbf{x}_i^\top$ (where $\sigma_i = \sigma(\mathbf{x}_i^\top \boldsymbol{\theta})$). This avoids the $O(nd^2)$ cost of forming the Hessian and the $O(d^3)$ cost of matrix inversion per iteration.
- It only requires first-order gradient information, making it easier to implement and more robust when the Hessian is ill-conditioned.
- It exhibits superlinear convergence, which is faster than gradient descent and often sufficient for practical purposes.

CONS:

- It converges more slowly than Newton's method, which has quadratic convergence near the optimum.
- It still requires storing a $d \times d$ matrix, leading to $O(d^2)$ memory usage, which becomes prohibitive for very high-dimensional problems (e.g., $d > 10^4$).
- For large-scale settings, the limited-memory variant L-BFGS is preferred, which stores only a few recent vectors instead of the full matrix.

1.3 In a logistic regression model, how would you interpret the coefficient β_i associated with a continuous predictor variable X_i ?

Holding all other predictors constant, a one-unit increase in X_i is associated with a change of β_i in the log-odds of the outcome being 1. Equivalently, the odds of the outcome multiply by a factor of e^{β_i} for each one-unit increase in X_i, assuming all other variables remain unchanged.

For example, if $\beta_i = 0.2$, then $e^{0.2} \approx 1.22$, meaning that a one-unit increase in X_i is associated with a 22% increase in the odds of the event occurring.

This interpretation concerns the odds, not the probability. The effect on the probability is non-linear and depends on the baseline value of the linear predictor.

1.4 Suppose you can access the `LogisticRegression()` function in python library. Write the pseudocodes to draw ROC curve of logistic regression on the training data D_{train}

```
X_train, y_train = D_train
model = LogisticRegression()
model.fit(X_train, y_train)

y_prob = model.predict_proba(X_train)[:, 1]

fpr, tpr, _ = roc_curve(y_train, y_prob)
auc_value = auc(fpr, tpr)

plot(fpr, tpr, label=f'ROC (AUC = {auc_value:.2f})')
xlabel('False Positive Rate')
ylabel('True Positive Rate')
title('ROC Curve on Training Data')
legend()
show()
```

P.S. This pseudocode assumes access to the `LogisticRegression()` class and standard evaluation/plotting functions such as `roc_curve`, `auc`, and `plot`. Actual implementation requires importing appropriate libraries or defining custom functions in Python.The plot function only gives a basic picture of the data without further beautification.

1.5 Explain the primary differences between LDA and logistic regression in classification

LDA and logistic regression differ in the following 5 aspects:
Modeling perspective:

- Logistic regression is a *discriminative* model. It directly models the conditional probability $P(Y = k \mid \mathbf{X} = \mathbf{x})$ using the logistic (or softmax) function.
- LDA is a *generative* model. It models the joint distribution $P(\mathbf{X}, Y)$ by assuming that the features $\mathbf{X}$ follow a multivariate Gaussian distribution within each class, i.e., $\mathbf{X} \mid Y = k \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma})$, with a common covariance matrix $\boldsymbol{\Sigma}$ across classes.

Assumptions:

- Logistic regression makes minimal assumptions—it only assumes the log-odds are linear in $\mathbf{x}$.

- LDA assumes: (1) Gaussian class-conditional distributions, (2) equal covariance matrices across classes, and (3) correctly specified prior class probabilities. Violations (e.g., non-Gaussian features or unequal covariances) can degrade LDA's performance.

Parameter estimation:

- Logistic regression estimates parameters by maximizing the conditional log-likelihood (via iterative methods like Newton–Raphson or BFGS).
- LDA estimates parameters (class means $\boldsymbol{\mu}_k$, shared covariance $\boldsymbol{\Sigma}$, and priors π_k) using closed-form sample statistics (e.g., empirical means and pooled covariance).

Robustness and efficiency:

- When LDA's assumptions hold, it is statistically more efficient than logistic regression (i.e., it achieves lower variance with fewer samples).
- When the assumptions are violated (e.g., features are not Gaussian), logistic regression is generally more robust because it does not rely on modeling the feature distribution.

Decision boundary:

- LDA's boundary depends on the estimated class priors and covariance structure.
- logistic regression's boundary is determined solely by the fit to the observed labels.

LDA is a generative model with strong distributional assumptions but high efficiency under those assumptions, whereas logistic regression is a flexible discriminative model that makes fewer assumptions and is more robust in practice, especially when the true data distribution deviates from Gaussianity.

1.6 Derive the discrimination function $\delta_k(x)$ in LDA using Bayes Theorem.

By Bayes' theorem, we classify $\mathbf{x}$ to the class k that maximizes the posterior probability:

$$P(Y = k \mid \mathbf{x}) \propto P(\mathbf{x} \mid Y = k)\,\pi_k,$$

where $\pi_k = P(Y = k)$ is the prior probability of class k.

In LDA, we assume $\mathbf{x} \mid Y = k \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma})$ with a common covariance matrix $\boldsymbol{\Sigma}$ for all classes. Thus,

$$P(\mathbf{x} \mid Y = k) \propto \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}_k)\right).$$

Taking the logarithm (which preserves the maximizer), we obtain:

$$\log P(Y = k \mid \mathbf{x}) \propto -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}_k) + \log \pi_k.$$

Expanding the quadratic form and discarding terms that do not depend on k ($-\frac{1}{2}\mathbf{x}^\top \boldsymbol{\Sigma}^{-1}\mathbf{x}$, since they do not affect the argmax), we define the discriminant function as:

$$\delta_k(\mathbf{x}) = \boldsymbol{\mu}_k^\top \boldsymbol{\Sigma}^{-1}\mathbf{x} - \frac{1}{2}\boldsymbol{\mu}_k^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k + \log \pi_k.$$

1.7 Show that the expected pooled sample covariance matrix

$$\widehat{\Sigma} := \frac{1}{n-K} \sum_{k=1}^{K} \sum_{i:y_i=k} (X_i - \widehat{\mu}_k)(X_i - \widehat{\mu}_k)^\top$$

in LDA is an unbiased estimator for the population covariance matrix Σ.

In LDA, assume there are K classes with total sample size n, where the k-th class contains n_k observations (so that $\sum_{k=1}^{K} n_k = n$), and all classes share the same population covariance matrix Σ. The pooled sample covariance matrix is defined as

$$\widehat{\Sigma} := \frac{1}{n-K} \sum_{k=1}^{K} \sum_{i:y_i=k} (X_i - \widehat{\mu}_k)(X_i - \widehat{\mu}_k)^\top,$$

where $\widehat{\mu}_k = \frac{1}{n_k} \sum_{i:y_i=k} X_i$ is the sample mean of class k. Observe that for each class k, the within-class sum of squares can be written as

$$\sum_{i:y_i=k} (X_i - \widehat{\mu}_k)(X_i - \widehat{\mu}_k)^\top = (n_k - 1) S_k,$$

where S_k is the sample covariance matrix of class k, i.e., $S_k = \frac{1}{n_k-1} \sum_{i:y_i=k} (X_i - \widehat{\mu}_k)(X_i - \widehat{\mu}_k)^\top$. Hence,

$$\widehat{\Sigma} = \frac{1}{n-K} \sum_{k=1}^{K} (n_k - 1) S_k.$$

Under the LDA model assumption, the observations in class k are i.i.d. from a multivariate normal distribution $\mathcal{N}(\mu_k, \Sigma)$, so for each k, the sample covariance matrix S_k is an unbiased estimator of Σ, i.e., $\mathbb{E}[S_k] = \Sigma$. Therefore,

$$\mathbb{E}[\widehat{\Sigma}] = \frac{1}{n-K} \sum_{k=1}^{K} (n_k - 1) \mathbb{E}[S_k] = \frac{1}{n-K} \sum_{k=1}^{K} (n_k - 1) \Sigma = \frac{1}{n-K} \left(\sum_{k=1}^{K} n_k - K \right) \Sigma = \frac{n-K}{n-K} \Sigma = \Sigma.$$

Thus, $\widehat{\Sigma}$ is an unbiased estimator of the population covariance matrix Σ.

1.8 Derive the discrimination function $\delta_k(x)$ in QDA using Bayes Theorem.

By Bayes' theorem, classify $\mathbf{x}$ to class k that maximizes:

$$P(Y = k \mid \mathbf{x}) \propto P(\mathbf{x} \mid Y = k)\, \pi_k.$$

In QDA, assume $\mathbf{x} \mid Y = k \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$, where each class has its own covariance matrix $\boldsymbol{\Sigma}_k$. Thus,

$$P(\mathbf{x} \mid Y = k) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}_k|^{1/2}} \exp\left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_k) \right).$$

Taking log and dropping constants independent of k ($-\frac{d}{2}\log(2\pi)$), the discriminant function is:

$$\delta_k(\mathbf{x}) = -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_k) - \frac{1}{2} \log |\boldsymbol{\Sigma}_k| + \log \pi_k.$$

Classification rule: $\hat{y} = \arg\max_k \delta_k(\mathbf{x})$. This is quadratic in $\mathbf{x}$, hence the name Quadratic Discriminant Analysis.

1.9 Explain how confusion matrix can be extended to multiclass classification problem? How about ROC?

For a multi-class classification problem with K classes, the confusion matrix generalizes to a $K \times K$ matrix. The entry in row i and column j represents the number of samples whose true class is i but were predicted as class j. The diagonal entries correspond to correct classifications, while off-diagonal entries indicate misclassifications between specific class pairs.

Extending ROC analysis to multi-class settings is less straightforward because ROC curves are inherently designed for binary classification. Common approaches include:

- **One-vs-Rest (OvR):** For each class k, treat it as the positive class and all other classes as the negative class. Compute the ROC curve and AUC for class k using its predicted probabilities. This yields K ROC curves and K AUC scores.

- **One-vs-One (OvO):** For every pair of classes (i, j), compute a binary ROC curve using only samples from these two classes. This results in $\binom{K}{2}$ ROC curves.

The overall performance is often summarized by averaging the AUCs (e.g., macro- or micro-averaged AUC), though interpretation depends on the chosen strategy.

1.10 Explain how the number of folds K affects the variance and bias of cross validation error

The choice of K in K-fold cross-validation involves a trade-off between bias and variance of the estimated test error.

When K is small , each training set is significantly smaller than the full dataset (using only $1 - \frac{1}{K}$ of the dataset). This leads to a less accurate model during each fold, causing the cross-validation error to overestimate the true test error—resulting in **higher bias**.

When K is large (e.g., $K = n$, i.e., leave-one-out cross-validation, LOOCV), each training set is nearly the full dataset, so the models are almost unbiased. However, the validation sets across folds overlap greatly, making the error estimates highly correlated. This correlation *increases the variance* of the overall cross-validation error estimate.

Thus, small K yields **higher bias but lower variance**, while large K yields **lower bias but higher variance**.

2 Predicting Breast Cancer

2.1 All-used LR model and ROC Curve

According to the modeling protocol, a logistic regression model was fitted to classify breast cancer cases as benign or malignant. Prior to model fitting, variance inflation factor (VIF) analysis revealed severe multicollinearity between Cell.size and Cell.shape (VIF $>$ 10). To ensure model stability and interpretability, Cell.size was removed, and the final model was estimated using the remaining predictors. The estimation results are summarized in Table 2.

Table 1: Variance Inflation Factors (VIF) for Predictor Variables

No.	Predictor	VIF
1	Cell.shape	13.33
2	Cell.size	13.26
3	Bl.cromatin	8.01
4	Epith.c.size	6.42
5	Cl.thickness	5.33
6	Bare.nuclei	5.29
7	Normal.nucleoli	5.20
8	Marg.adhesion	4.80
9	Mitoses	2.70

Table 2: Logistic Regression Results (After Removing `Cell.size`)

Predictor	Coef.	Std. Err.	z	$P > \|z\|$	95% CI	
					[0.025	0.975]
const	-1.3969	0.492	-2.837	0.005	-2.362	-0.432
Cl.thickness	1.1198	0.606	1.848	0.065	-0.068	2.307
Cell.shape	0.9814	0.630	1.558	0.119	-0.253	2.216
Marg.adhesion	0.9312	0.395	2.356	0.018	0.157	1.706
Epith.c.size	0.9098	0.586	1.551	0.121	-0.240	2.059
Bare.nuclei	1.0269	0.436	2.356	0.018	0.172	1.881
Bl.cromatin	1.4111	0.651	2.168	0.030	0.136	2.687
Normal.nucleoli	0.4951	0.511	0.969	0.333	-0.506	1.497
Mitoses	0.5709	0.788	0.725	0.469	-0.973	2.115

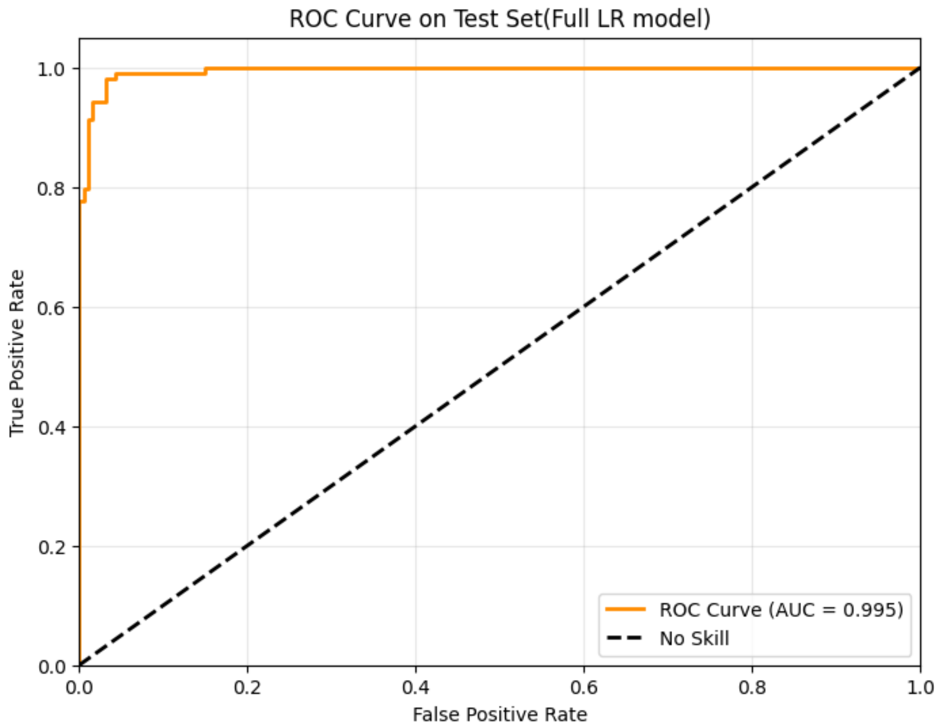

Figure 1: ROC curve on the test set for the specific LR model

2.2 Partly-used LR model and ROC Curve

Using almost the same method as problem 2.1. The results are shown in Table 3 and Figure 2

Table 3: Logistic Regression Results (Using Selected Predictors)

Predictor	Coef.	Std. Err.	z	$P > \|z\|$	95% CI	
					[0.025	0.975]
const	-1.3602	0.392	-3.470	0.001	-2.128	-0.592
Cl.thickness	1.3730	0.546	2.513	0.012	0.302	2.444
Cell.shape	1.4800	0.603	2.455	0.014	0.298	2.662
Marg.adhesion	1.0303	0.362	2.844	0.004	0.320	1.740
Bare.nuclei	1.0734	0.397	2.707	0.007	0.296	1.851
Bl.cromatin	1.9602	0.592	3.314	0.001	0.801	3.120

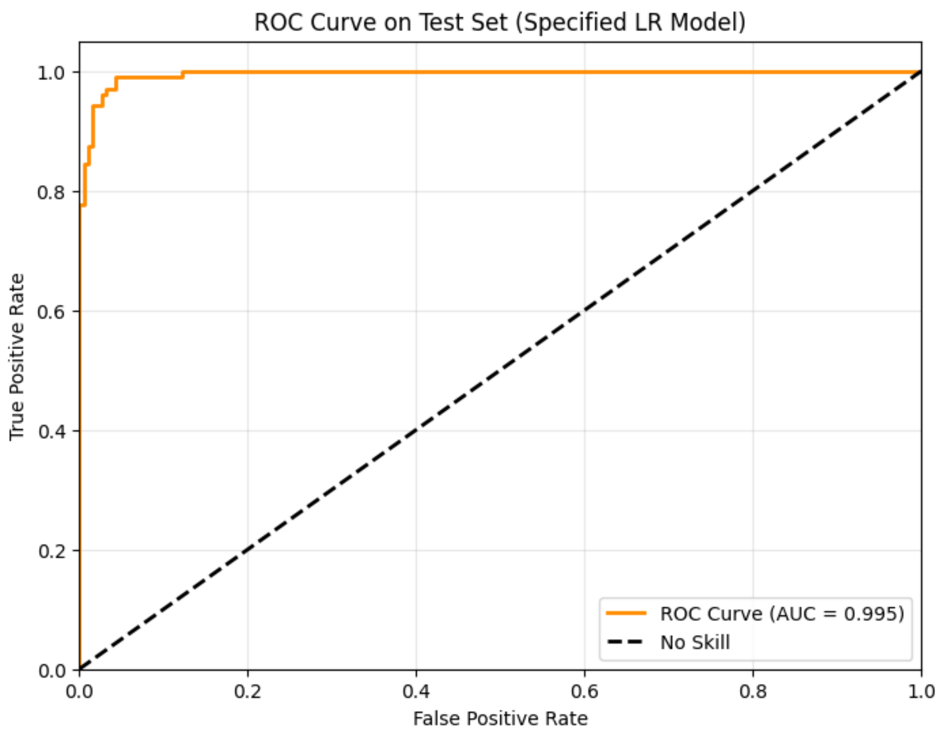

Figure 2: ROC curve on the test set for the specific LR model

2.3 All-used LDA model and ROC Curve

The LDA discriminant coefficients indicate the contribution of each standardized predictor to the separation of benign and malignant tumors. Bare.nuclei exhibits the strongest positive association with malignancy (coefficient = 4.62), followed by Cl.thickness (2.20) and Cell.shape (1.73). All variables except Mitoses positively contribute to the malignant class.

Table 4: LDA Discriminant Coefficients (Standardized Predictors)

Predictor	LDA Coefficient
Bare.nuclei	4.6232
Cl.thickness	2.2027
Cell.shape	1.7298
Normal.nucleoli	1.6294
Bl.cromatin	1.5707
Epith.c.size	1.4600
Marg.adhesion	1.2804
Cell.size	1.2105
Mitoses	-0.2556

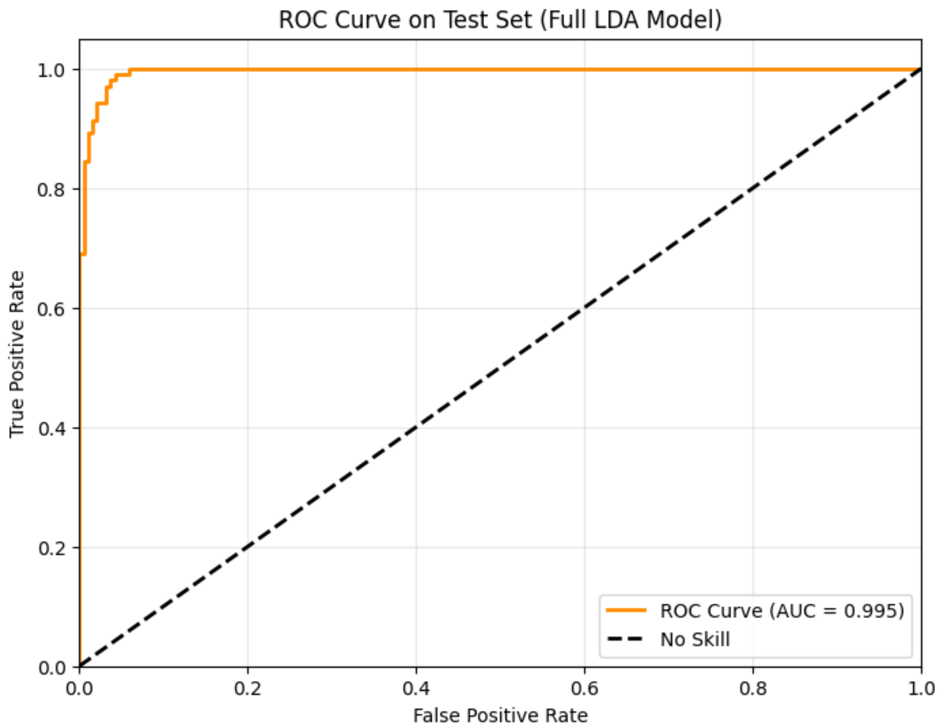

Figure 3: ROC curve on the test set for the full LDA model

2.4 Partly-used LDA model and ROC Curve

The specified LDA model identifies Bare.nuclei as the strongest discriminator of malignancy (coefficient = 4.23), followed by Cell.shape (3.38) and Bl.cromatin (2.34). These nuclear and morphological features dominate the discriminant direction.

Table 5: LDA Discriminant Coefficients (Standardized Predictors)

Predictor	LDA Coefficient
Bare.nuclei	4.2297
Cell.shape	3.3823
Bl.cromatin	2.3411
Cl.thickness	2.1741
Marg.adhesion	1.5543

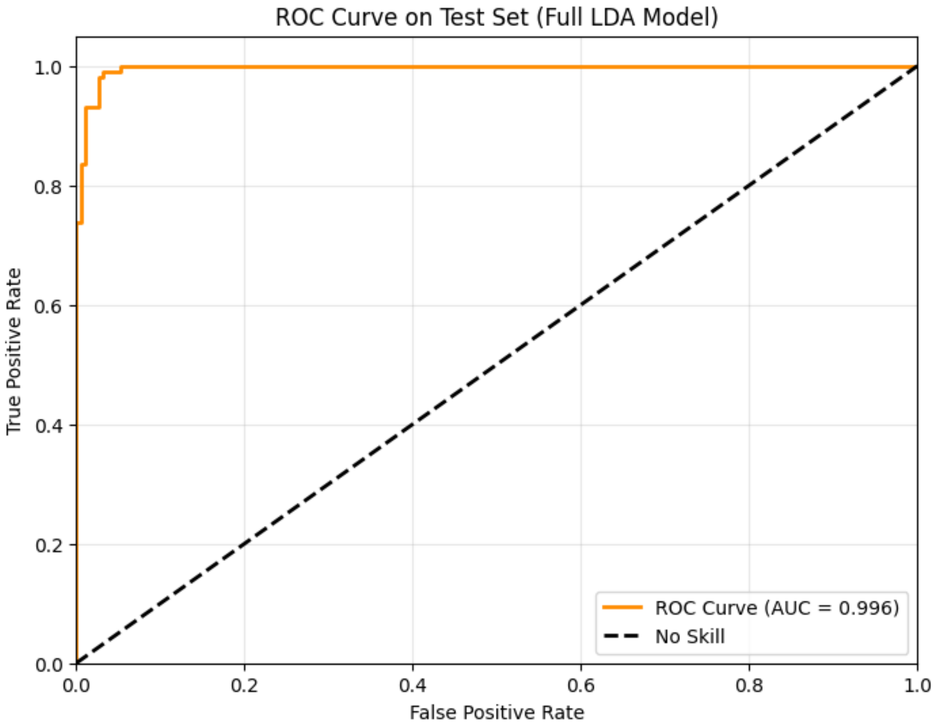

Figure 4: ROC curve on the test set for the specified LDA model

2.5 All-used QDA model and ROC Curve

Unlike LDA, Quadratic Discriminant Analysis (QDA) does not produce a single set of discriminant coefficients. This is because QDA estimates a separate covariance matrix for each class and uses a quadratic decision boundary. Consequently, QDA does not provide global feature weights analogous to the `coef_` attribute in logistic regression or LDA, and therefore cannot be summarized by a simple list of "coefficients" for each predictor variable.

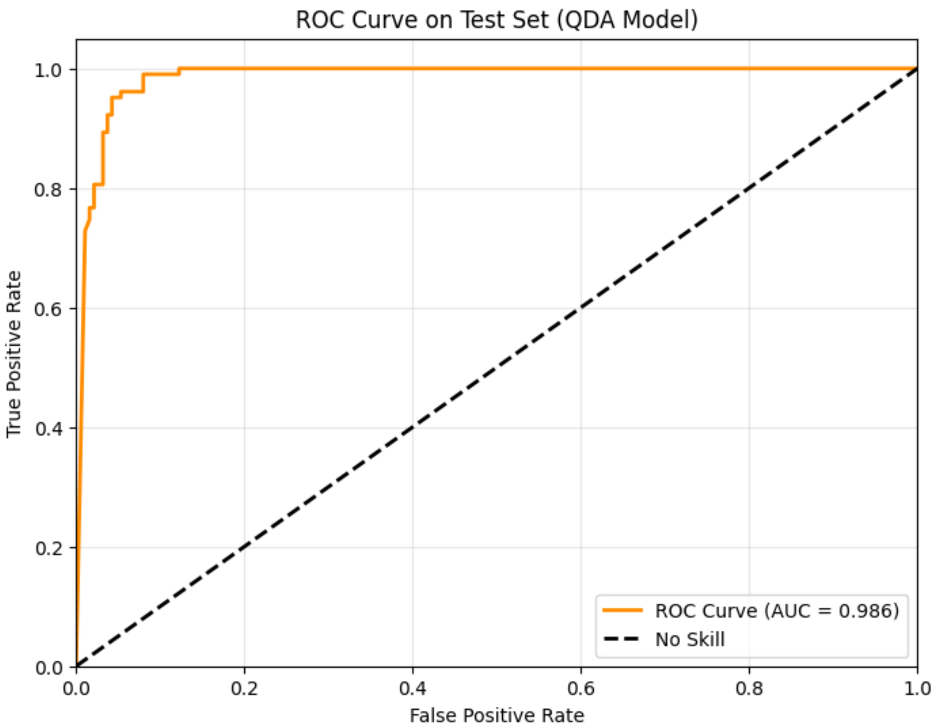

Figure 5: ROC curve on the test set for the full QDA model

2.6 Compare LR, LDA and QDA models by AUC values

Table 6: Comparison of Model Performance on Test Set (AUC)

Model	Test Set AUC
Logistic regression (excluding `Cell.size`)	0.9951
Logistic regression (5 specific predictors)	0.9954
Full LDA (all predictors)	0.9955
LDA (5 specific predictors)	0.9963
Quadratic Discriminant Analysis (QDA)	0.9856

All five models demonstrate excellent classification performance, with AUC values above 0.985. The best-performing model is LDA using the 5 specified predictors (0.9963), suggesting that a parsimonious feature set combined with the linear Gaussian assumption yields optimal discriminative power for this dataset. In contrast, QDA achieves the lowest AUC (0.9856), indicating that the more flexible quadratic decision boundary may lead to overfitting given the sample size and feature correlations.

3 Speed and Stopping Distances of Cars

3.1 LOOCV errors vs. polynomial degrees

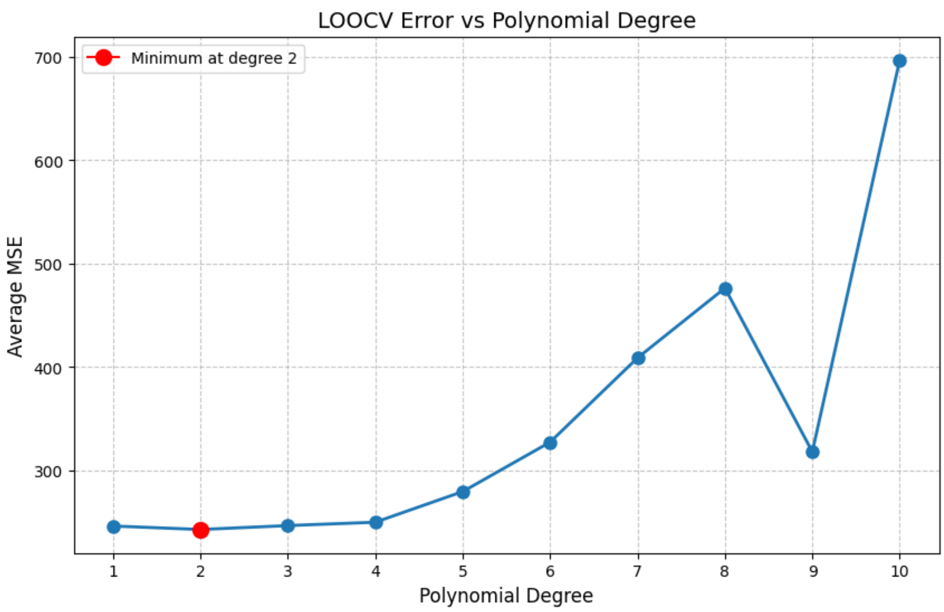

Figure 6: LOOCV errors vs. polynomial degrees

From the LOOCV error plot, the mean squared error is minimized at polynomial degree 2 , indicating a quadratic model best captures the speed–distance relationship. Although there's dip at degree 9 ,but it's likely an artifact of high-variance overfitting — with only 50 data points, higher-degree polynomials can fit noise, causing unstable CV errors,which should not be interpreted as meaningful improvement.

3.2 K-Folds CV errors vs. polynomial degrees at $K = 5$

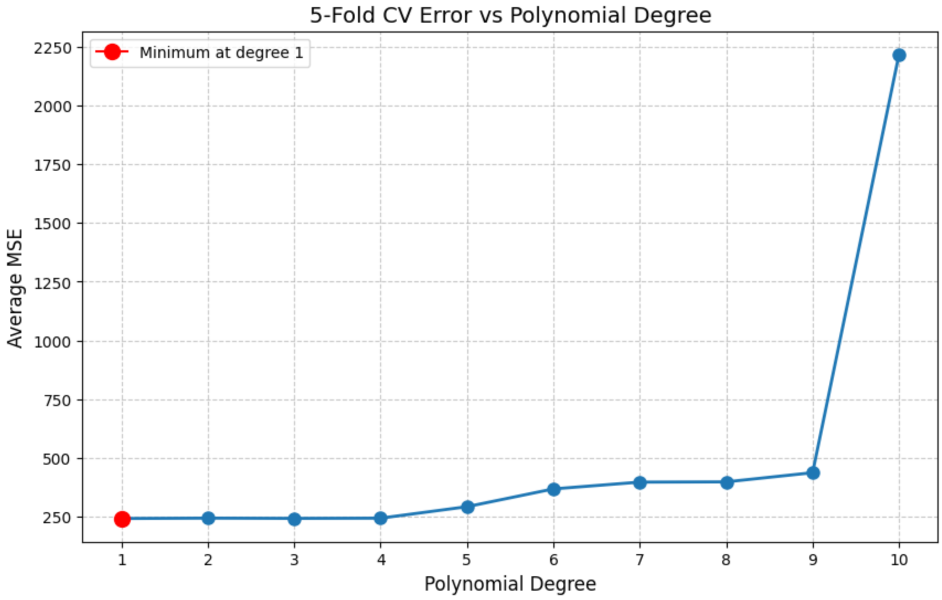

Figure 7: 5-Fold CV errors vs. polynomial degrees

The 5-fold CV error plot shows the minimum average MSE occurs at degree 1, suggesting a linear model provides the best generalization performance. Higher-degree polynomials (≥ 2) exhibit increasing or stable error, indicating no significant improvement from added complexity — and severe overfitting at degree 10.

3.3 KNN-Gaussian bandwidth CV optimization diagram

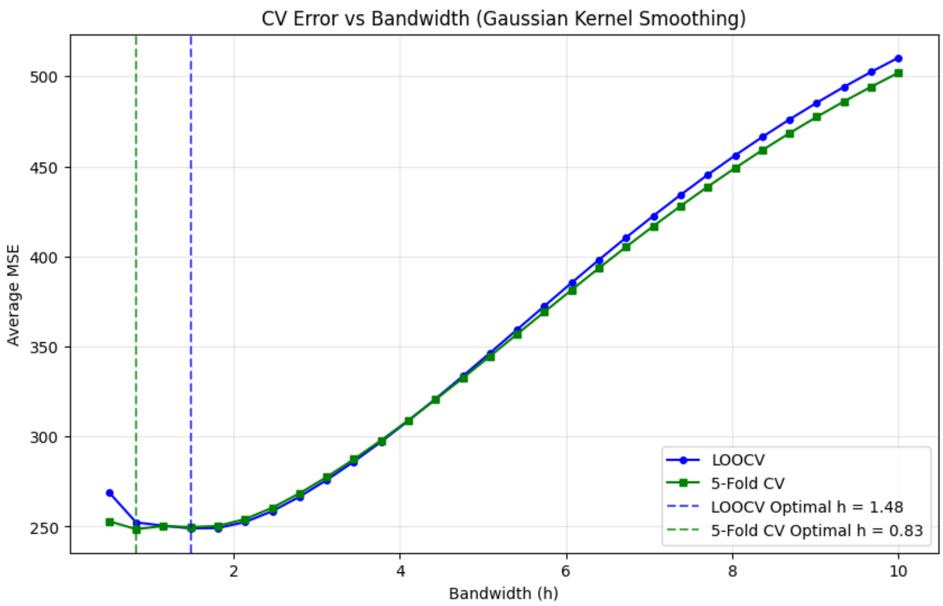

Figure 8: KNN-Gaussian bandwidth CV optimization diagram

The plot shows that both LOOCV (blue) and 5-Fold CV (green) errors decrease initially as bandwidth h increases from 0.5, then rise steadily for larger h. The optimal bandwidths are $h = 1.48$ (LOOCV) and $h = 0.83$ (5-Fold CV), indicating a small-to-moderate smoothing level is best. The slight difference between the two suggests 5-Fold CV may be more conservative in selecting smaller bandwidths. Both methods agree that overly large bandwidths ($h > 4$) cause underfitting (high bias), while very small h leads to overfitting. For this dataset, Gaussian kernel smoothing with $h \approx 1$ provides the best trade-off.

3.4 Compare KNN-Gaussian and polynomial regression

Polynomial regression (degree 2) and Gaussian kernel smoothing ($h \approx 1$) both fit the speed–distance relationship well, but differ in approach. The quadratic polynomial is simpler, more stable, and physically interpretable (distance $\propto$ speed^2). Kernel smoothing is flexible but less stable and offers no advantage here since the true trend is nearly quadratic. Thus, polynomial regression is preferred for this dataset.

4 Titanic

4.1 coef and 95% CI in Linear Regression

Table 7: Logistic Regression Coefficients and 95% Confidence Intervals

Variable (Reference)	Estimate ($\hat{\beta}$)	95% CI
Sex = male (vs. female)	-2.6161	$[-3.0383,\ -2.1939]$
Pclass = 3rd (vs. 1st)	-2.5258	$[-3.1816,\ -1.8700]$

4.2 coef and 95% CI in Linear Regression

Table 8: Estimated Coefficients and 95% Confidence Intervals: Wald vs. Bootstrap

Variable (Reference)	Estimate	Wald 95% CI	Bootstrap 95% CI
Sex = male (vs. female)	-2.6161	$[-3.0383,\ -2.1939]$	$[-3.0819,\ -2.2201]$
Pclass = 3rd (vs. 1st)	-2.5258	$[-3.1816,\ -1.8700]$	$[-3.2251,\ -1.9044]$

The bootstrap confidence intervals are very close to the Wald intervals reported earlier. In both cases, the bootstrap intervals are slightly wider and shifted modestly downward, but the differences are minimal. This close agreement suggests that the asymptotic assumptions underlying the Wald intervals (e.g., large-sample normality of the MLE) are reasonably well satisfied in this dataset. The bootstrap results thus corroborate the validity of the standard logistic regression inference.

4.3 Explore the dataset

- The overall survival rate was approximately 38%.
- Survival differed dramatically by gender: about 74% of females survived, compared to only 19% of males.
- Passenger class strongly influenced survival: first-class passengers had a survival rate of 63%, second-class 47%, and third-class just 24%.
- The average age was 29.7 years (range: 0.42 to 80 years), with children showing higher survival probability.
- The mean fare was $32.20, and higher fares (associated with higher class) correlated with greater survival chances.

Table 9: Exploration on Titanic dataset

Statistic	**Value**
Total passengers	891
Overall survival rate	38.4%
Female survival rate	74.2%
Male survival rate	18.9%
First-class survival rate	63.0%
Second-class survival rate	47.3%
Third-class survival rate	24.2%
Mean age (years)	29.7
Age range (years)	0.42 – 80.0
Mean fare (USD)	$32.20

4.4 Predicting Survival Probability for a Single Passenger with Bootstrap Uncertainty Quantification

Table 10: Predicted Survival Probability for the First Passenger (Test Point)

Quantity	Value
Predicted survival probability	0.0896
95% Bootstrap prediction interval	[0.0578, 0.1263]

4.5 Predicting Survival Probability for a Single Passenger Using QDA with Bootstrap Uncertainty Quantification

Table 11: Predicted Survival Probability for the First Passenger: LR vs. QDA

Model	Predicted Probability	95% Bootstrap Prediction Interval
Logistic Regression (LR)	0.0896	[0.0578, 0.1263]
Quadratic Discriminant Analysis (QDA)	0.0371	[0.0176, 0.0687]

The predicted probability from QDA is close to 0, and the range is very narrow, indicating that the model is highly confident that this passenger (a 22-year-old male in 3rd-class, with a low fare) will not survive, which fits the truth. Compared to QDA, LR's prediction is slightly more optimistic, but still falls within the low probability range.